Sebastian Parakal Jose

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Summary

Software engineer with **3+ years** of experience building backend services, cloud systems and mobile apps that people rely on. Two years at NCR / Global Payments on a 24/7 payments platform for **Mastercard** and **American Express**, working across teams in **C++**, **Java (Spring Boot)**, **Node.js (Express, NestJS)**, **PHP (Laravel)**, **Go** and **.NET**. As CTO of MOSAIC, directed the launch of the IFRC World Disasters Report platform to **30,000+** users on day one. Builds on **AWS**, **Kafka** and **Kubernetes**, ships mobile apps in **React Native**, Flutter and Kotlin, and designs multi-agent **Claude Code** workflows with independent verification. Master of IT, Monash (**GPA 4.0/4.0**). **Unlimited work rights in Australia.**

Education

Monash University — Melbourne, Australia

Master of Information Technology – ***GPA 4.0/4.0 (WAM 84.867)*** — *Jul 2024 – Sep 2026*

- High Distinctions in Cloud Computing and Security, Mobile and Distributed Computing Systems, Advanced Database Technology, Internet Applications Development, Software Engineering and Foundations of Data Science

National Institute of Technology Calicut — Kerala, India

Bachelor of Technology in Computer Science and Engineering (First Class) — *2018 – 2022*

Experience

Chief Technology Officer (CTO) — Jan 2026 – Aug 2026

Monash Students for AI with Communities (MOSAIC) — *Melbourne, Australia*

- Directed the production launch of the **IFRC** World Disasters Report platform (wdr26.org), launched globally in Geneva on 5 March and serving **30,000+** users on day one
- Top-three contributor (**110+ commits**) in **Next.js** and **React**: newsletter sign-up via the Brevo SDK and Next.js server actions (with tests), an in-app bug-report flow, Arabic (right-to-left), Russian and French localisation, and launch-day UI
- Contributed to **Ctrl+Alt+Prebunk**, an IFRC game that trains players to spot misinformation, built in **Next.js**, **React** and **TypeScript** with **Zustand** state and Jest/React Testing Library tests: feed interactions, persistence fixes and large-screen layout
- Led development of and built features for **Disinformer**, a cross-platform **Flutter** (Dart) misinformation game for IFRC National Societies on iOS, Android and web, backed by **Firebase**, with offline sync, localisation and analytics
- Rolled out AI-assisted development (**Claude Code**, GitHub Copilot, Cursor), code review and Notion onboarding playbooks, so new joiners shipped in their first week

Teaching Associate — Jul 2025 – Present

Monash University — *Melbourne, Australia*

- Mentor **100+** students in Android development (**Kotlin**, Jetpack Compose, MVVM, REST APIs); rated **91%** for satisfaction (SETU)
- Built a multi-agent **Claude Code** marking harness: an orchestrator hands each submission to five sub-agents (code-inspector, report-auditor, rubric-scorer, uniqueness-checker and an independent **verifier**) that flags hallucinations, wrong files and arithmetic errors, with a full audit log
- Built **Python** tooling around the agents for anything countable: PPTX/PDF/DOCX extraction, reference checks against the Semantic Scholar and OpenAlex **APIs**, and a validation gate on marks; workbooks are backed up before every write

Software Engineer — Jul 2022 – Jul 2024

NCR Corporation / Global Payments — *Hyderabad, India*

- Engineered mission-critical **C++** backend services for a 24/7 payment processor on Linux, handling **Mastercard** and **American Express** transactions across ATM, payment-switch and card-payment flows
- Built services and **REST APIs** across multiple teams and products in **Java** (**Spring Boot**), **Node.js** (**Express**, **NestJS**, TypeScript) and **PHP** (**Laravel**)
- Implemented and extended **ISO 8583** messaging pipelines routing high-volume transactions between card networks, acquirers and issuing banks, without disrupting live integrations

- On the **L4 escalation team**, root-caused the hardest production incidents and shipped hotfixes under live constraints, improving system reliability by **20%**
- Built **Splunk** dashboards, queries and alerts for pipeline observability, cutting mean time to detect anomalies by **30%**
- Tuned **Oracle SQL** queries, indexes and stored procedures behind transaction records, cutting peak-load latency by **25%**
- Built a **.NET 6** automation framework (SpecFlow, MSTest) for EMV transaction flows with **Jira** integration, containerised with **Docker** on Oracle Linux and run in **GitHub Actions** and **Jenkins**
- Rewrote legacy **Perl** scripts as maintainable **Go** tools, and built .NET utilities for encryption and **DUKPT** key management

Projects

Hush | hushapp.live | *Vue 3, AWS Lambda, API Gateway, RDS PostgreSQL/PostGIS, S3, EventBridge, Terraform*

- Sole engineer on a live app for sensory-sensitive people: **Vue 3** and Mapbox front end, **Python** Lambdas behind **API Gateway**, and fastest and quietest walking routes computed in **PostgreSQL** on **RDS** with PostGIS and pgRouting
- Chose to **precompute** an hourly scikit-learn crowd forecast in a container Lambda and publish it to **S3**, instead of running the model on demand, so cost stays flat and the map never waits on a model
- All infrastructure in **Terraform** with a least-privilege IAM role per function; **GitHub Actions** runs Vitest unit and Playwright E2E tests on every PR; shipped an OpenAPI spec, runbook and risk register for handover

SignalStack | *Python, Apache Kafka, PostgreSQL, Redis, FastAPI, Next.js, Terraform, Prometheus, Grafana*

- Built a self-hosted tech-trend platform ingesting real-time activity from **GitHub** and **Hacker News** through an event-driven **Apache Kafka (KRaft)** pipeline on a single free-tier AWS EC2 instance
- Fault-tolerant consumers with partitioned topics, batched **PostgreSQL** writes and **Redis** deduplication, raising throughput ~**250x** at zero added infrastructure cost
- Served through a 9-endpoint **FastAPI** service with Redis caching and a **Next.js** front end; provisioned with **Terraform**, CI/CD in GitHub Actions, monitored with self-hosted **Prometheus** and **Grafana**

CloudPose | *Python, FastAPI, Docker, Kubernetes, YOLOv8, Locust*

- Profiled and optimised the inference pipeline (image resizing, explicit garbage collection, concurrency control), **cutting memory 35%** and **doubling throughput** across horizontally scaled pods

BirdTag | *AWS Lambda, API Gateway, DynamoDB, S3, Cognito, SNS, ECR*

- Serverless media library that auto-tags bird species in uploaded photos, video and audio (YOLO for images and video, BirdNET for audio, shipped as Lambda layers and **ECR** images) with **SNS** species alerts

FridgeFolks | *React Native, iOS App Store*

- Built and published **FridgeFolks**, a **React Native** app on the **iOS App Store** that cuts household food waste: barcode scanning, expiry-date tracking, shared pantries and reminder notifications

Technical Skills

Languages: Python, Java, Kotlin, TypeScript, JavaScript, C++, C#, Go, PHP, Dart, SQL, Bash, R
Backend: Spring Boot, Node.js (Express, NestJS), Laravel, .NET, FastAPI, Flask, REST API design (OpenAPI), Apache Kafka, ISO 8583, microservices
Cloud & Infrastructure: AWS (Lambda, API Gateway, S3, DynamoDB, RDS, EC2, ECR, Cognito, EventBridge, SNS, Amplify), Cloudflare Workers and D1, Docker, Kubernetes, Terraform
Data: PostgreSQL (PostGIS, pgRouting), Oracle SQL, DynamoDB, Redis, Firebase/Firestore, Room, star-schema data warehousing
Frontend & Mobile: React, Next.js, Vue 3, Zustand, Tailwind CSS, React Native, Flutter, Android (Jetpack Compose, MVVM, WorkManager, ML Kit)
Quality & Delivery: GitHub Actions, Jenkins, CodeQL, Dependabot, SonarQube; Jest, React Testing Library, Vitest, Playwright, pytest, MSTest, SpecFlow, Locust
Observability & Security: Splunk, Prometheus, Grafana, CloudWatch; least-privilege IAM, secrets management, CORS and security headers, DUKPT
AI Engineering: Claude Code, sub-agents and multi-agent orchestration, harness engineering, GitHub Copilot, Cursor, YOLO, ONNX Runtime, scikit-learn